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/**
 * Array
 * -----
 *
 * This short macro showcases different ways to use variable (string and number)
 *
 * Due to the simple nature of this code, no copyright is applicable
 *
 * Code created for Image Processing and Analysis For Life Scientists MOOC on EdX
 * https://www.edx.org/course/image-processing-and-analysis-for-life-scientists
 *
 * 2018 - Romain Guiet, EPFL - SV - BIOP
 * https://biop.epfl.ch
 */

// We start by closing all the images
run("Close All");
// and Clear the log window.
print("\Clear");

// We define a new array, with elements (here, possible LookUpTable) separated by commas.
color_array = newArray( "Red", "Green", "Blue", "Gray", "Fire" );

// Let's open "blobs.gif"
run("Blobs (25K)");
// and get its ID (so we can re-select this image when needed)
blobs_id = getImageID();
// a comment in a dialog window.
print ("color_array : ");
Array.print(color_array);
selectWindow("Log");
waitForUser("We'll change the LookUp Table of the image to Red,\nusing the variables stored in the 'color_array'\n(you can have a look to the Log window)");

// Change the LUT
//
// make sure to select the image
selectImage(blobs_id);
// To Take the FIRST element , we use the index 0
run(color_array[0]);

waitForUser("Now, we'll change the LookUp Table of the image to the last element of the 'color_array'");
// make sure to select the image
selectImage(blobs_id);
// To Take the Last element , we use the lengthOf(color_array)-1 (because indexes start at 0)
run(color_array[lengthOf(color_array)-1]);

// reorganize images on screen
run("Tile");

print ("=====");
waitForUser("Next, we'll create a line on the image,\nand measure pixel values among that line");
//Here, we make the line to between defines coordinates makeLine(x1,y1,x2,y2)
makeLine(120, 30, 160, 85);
// store the measurements in an array 'lineProfile'
lineProfile_array = getProfile();
// Print to the Log window
print ("Intensity profile: ");
Array.print(lineProfile_array);
selectWindow("Log");

// You may have notice by now that to print array we can not use function print(...)
// but we have to use Array.print(array) instead.
// There is other functions that you can use with the array (cf. https://imagej.net/ij/macros/functions)

Array.getStatistics(lineProfile_array, min, max, mean, stdDev);
print ("min value of 'lineProfile_array' :"+ min);
print ("max value of 'lineProfile_array' :"+ max);

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// You can find on imagej.net an example script  
// that make use of array functions  
// https://imagej.net/ij/macros/examples/ArrayFunctions.txt
```